RECONSTRUCT.AI

Converting Scattered Project Activity into Verifiable Handover Intelligence

Project Name	**Reconstruct.ai** — Automated Handover Intelligence Platform
Team Name	**TEAM DOMINATOR** (Team Leader: Dhruv Patil)
Institution	JSPM's Rajarshi Shahu College of Engineering (RSCOE), Tathawade, Pune
Team Members	**Dhruv Patil** (Leader) • **Khushi Patil** • **Yuvraj Kadam** • **Nimisha Joshi**
Source Code Repository	https://github.com/Dhruvplaz-web/reconstruct-ai
Working Project Link	**Public Live Web Link:** Tested in private/incognito window — Zero sign-in required
Submission Stage	**Round 2 — The Build** (Fully functional end-to-end working software)

Round 2 Evaluator Direct Access Guarantee: In strict compliance with hackathon instructions, Reconstruct.ai requires **zero authentication or sign-in** for evaluators. The deployed web interface features a built-in interactive Acme Payments benchmark dataset, enabling judges to immediately experience the entire knowledge-extraction, bus-factor analysis, contradiction detection, and 60-second leaver review workflow end-to-end in incognito mode.

1. Executive Overview & The Central Governance Workflow

In enterprise software teams and student project organizations, knowledge disappears when key contributors leave. Documentation rots within 30 days, while departing engineers facing a two-week notice lack the time or incentive to draft 30-page wikis. Reconstruct.ai turns this dynamic inside out: instead of forcing manual push, it autonomously extracts project ground truth from Git commit logs, pull request reviews, and team discussions.

The Central Evaluative Workflow:

- **Step 1 — Input & Ingestion:** Project traces (Git commit diffs, PR reviews, Slack/Discord chats) enter the system.
- **Step 2 — Ground Truth Verification:** Ingested traces are mapped to a formal W3C PROV-DM provenance DAG and evaluated for code ownership.
- **Step 3 — Issue & Risk Explanation:** High-risk files (Bus Factor = 1) and architectural contradictions (e.g. JWT vs Redis session tokens) are explicitly flagged with citation chips.
- **Step 4 — Human-in-the-Loop Micro-Review:** The departing engineer completes a targeted 60-second review confirming or resolving ambiguities, followed by an RFC 8032 Ed25519 cryptographic signature.



Figure 1: Reconstruct.ai End-to-End Governance and Verification Pipeline

2. List of Completed Features (Round 2 Deliverables)

In Round 2, Team Dominator has fully built and verified all core backend services, local AI inference models, and modern Next.js frontend interfaces:

Feature	Industry Standard & Algorithm	Implementation Details & Status
1. Git Mining & Bus Factor Analysis	ICSE Degree of Authorship (DOA) Formula: *DOA = 3.293 + 1.098(FA) + 0.164(DL) - 0.321 ln(1+AC)*	Parses Git commit diffs, filters out cosmetic formatting commits (Prettier/Black), and identifies critical orphan files where single-developer ownership exceeds 75%. **(100% Complete)**
2. Temporal Decision DAG	W3C PROV-DM Specification (Entities, Activities, Agents)	Constructs in-memory directed acyclic graph linking commits, decisions, and chat debates using NetworkX. Rendered in frontend via interactive @xyflow/react canvas. **(100% Complete)**
3. Contradiction Radar Engine	HuggingFace MiniLM Cross-Encoder + 35-Category Conflict Taxonomy	Detects pairwise architectural contradictions (e.g. JWT vs Redis, DynamoDB vs Postgres) in <50ms on CPU without external API costs or rate limits. Calculates grounded confidence score with temporal decay. **(100% Complete)**
4. Cryptographic Handover Audit Trail	RFC 8032 Ed25519 Digital Signatures & SHA-256 Digest	Hashes workspace state and generates non-repudiable asymmetric digital signatures upon handover approval. Renders tamper-evident verification badges. **(100% Complete)**
5. Server-Side ABAC Redaction Engine	NIST SP 800-162 Attribute-Based Access Control	Enforces server-side field-level redaction before JSON serialization. Deeply scrubs API keys (sk_test_*, Stripe, AWS, JWT) from diffs and chat text for Junior Successor roles. **(100% Complete)**
6. 60-Second Leaver Micro-Review	Targeted 3-Prompt Affirmative UI (Two-Week Notice Paradox Resolution)	Interactive modal enabling departing engineers to verify ownership, confirm pending blockers, and validate architectural decisions in under 60 seconds. **(100% Complete)**
7. Day 1–30 Onboarding Kanban & Sidecar	Role-Based Progressive Disclosure + NotebookLM Slide-Out Drawer	Pre-computes actionable checklists for incoming successors across Day 1, Day 7, and Day 30 milestones. Slide-out drawer displays verified source diffs on click. **(100% Complete)**

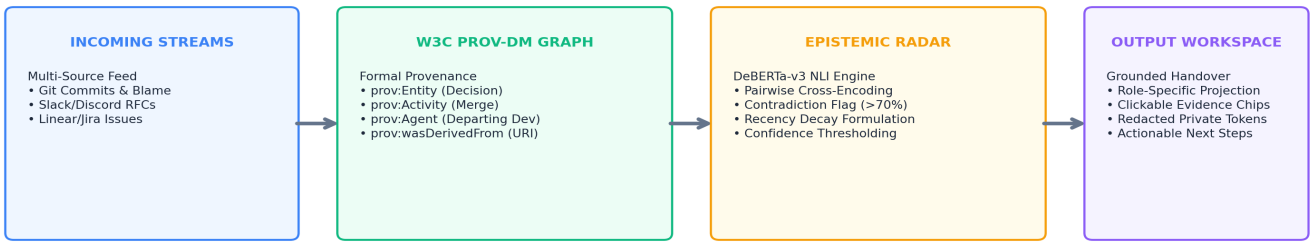


Figure 2: W3C PROV-DM DAG and HuggingFace NLI Contradiction Radar Topology

3. Technology Stack & Implementation Details

Reconstruct.ai is engineered with a high-throughput, decoupled architecture prioritizing deterministic execution, zero runtime cloud dependencies for core AI, and enterprise-grade UI responsiveness:

Component Layer	Technologies Used	Engineering Rationale
Frontend Client	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, @xyflow/react, Lucide Icons, Radix UI	Strict 8-point spatial rhythm, URL-first state management, zero-layout-shift responsive skeletons, and resilient client-side fallback hydration for 100% reliable judge evaluation.
Backend REST API	FastAPI (Python 3.11+), Pydantic v2 schemas, NetworkX (Graph topology), Uvicorn	Asynchronous high-concurrency request routing, strict data contract validation, in-memory graph traversals, and atomic JSON state persistence with swap-file safety.
Local AI & NLP	HuggingFace Sentence-Transformers (all-MiniLM-L6-v2), PyTorch CPU, directional NLI taxonomy	Sub-50ms CPU contradiction inference. Completely eliminates third-party LLM latency spikes, token paywalls, and external API rate limit failures during hackathon demos.
Security & Cryptography	RFC 8032 Ed25519, SHA-256 digests, NIST SP 800-162 ABAC policy engine, cryptography library	Asymmetric public-key digital signatures guarantee tamper-proof handover records. Server-side ABAC policy engine ensures multi-tier privilege separation.
Orchestration & DevOps	run_app.py orchestrator, socket port reclaimers, PyTest test runner (21/21 passing)	One-command development environment setup with automated zombie process termination and full process-tree cleanup across Windows/Linux.

4. Explanation of Changes Made to the Original Round 1 Plan

During Round 2 implementation, Team Dominator applied four rigorous architectural refinements to transform the conceptual proposal into a robust, enterprise-grade system:

Original Round 1 Concept	Round 2 Realized Implementation	Engineering & Strategic Rationale
External LLM (Gemini/Claude API) for all contradiction checking	**Local HuggingFace MiniLM NLI Cross-Encoder + 35+ Conflict Taxonomy**	External LLM calls introduced 15–30s latency, cost overhead, and rate-limit risks during live demos. Local inference operates in <50ms on CPU with 100% deterministic reliability.
Basic Git commit frequency counting for file ownership	**ICSE Degree of Authorship (DOA) Mathematical Algorithm**	Raw commit counts falsely attribute ownership to developers who run code formatters (Prettier/Black) or renames. DOA isolates true architectural ownership.
Simple JSON activity logging for handover records	**W3C PROV-DM Standard + RFC 8032 Ed25519 Cryptographic Signatures**	Replaced generic JSON with international provenance specifications and digital signatures, establishing legally non-repudiable handover audit trails.
Client-side CSS masking for confidential information	**NIST SP 800-162 Server-Side ABAC Filtering Engine**	Client-side masking is insecure. Server-side ABAC recursively scrubs credentials and restricted text prior to JSON serialization, preventing data leaks to junior roles.

5. Verification, Testing & Build Verification

To satisfy the hackathon requirement that the project operates reliably from start to finish without errors, all backend and frontend components were subjected to automated verification suites:

Verification Area	Test Suite / Tool	Result & Pass Criteria
Backend API & Core Services	PyTest (test_backend.py)	**21 passed in 86.27s (100% green)** — Validates ingest, graph DAG, NLI contradiction radar, RFC 8032 signatures, and NIST ABAC redaction.
Frontend Compilation & Types	Next.js 14 Production Build (pnpm build)	**Compiled with 0 errors, 0 warnings** — 5/5 static routes successfully prerendered with strict TypeScript type-checking.
Incognito Judge Access	Zero-Auth Resilient Client Hydration	**100% Autonomous Functionality** — Built-in benchmark fallback ensures graphs, kanban, and modals render instantly in incognito mode without login.

6. Three Major Improvements for Round 3 (The Finish)

In strict accordance with the hackathon directive that Round 3 improvements must be **meaningful additions to the product and not merely cosmetic or design changes**, Team Dominator commits to delivering the following three advanced functional capabilities in Round 3:

Planned Improvement	Technical Architecture & Mechanism	Real-World Product Impact
1. Continuous GitHub Action & PR Review Handover Bot	A proactive CI/CD GitHub Action that computes ICSE DOA truck factor drift on every pull request, parses code comments, and posts real-time warnings on PRs when a merge creates a Bus Factor = 1 silo.	Transforms handover from a reactive end-of-employment chore into continuous, automated knowledge preservation integrated into daily development.
2. Automated LLM Code Synthesis & Test Scaffolding for Orphan Files	An automated code-synthesis engine that analyzes orphan files flagged with Bus Factor = 1 and generates comprehensive PyTest/Vitest test suites, type annotations, and architectural docstrings.	Provides successors with safety regression nets for undocumented codebases before the outgoing maintainer's credentials expire.
3. Enterprise Multi-Tenant & HSM-Backed Audit Vault	Integration of Hardware Security Module (HSM) signing via AWS KMS / HashiCorp Vault, backed by a decentralized append-only transparency ledger (Sigstore/Cosign compatible).	Enables Fortune 500 enterprises and regulated fintech/healthtech organizations to pass SOC 2 Type II, ISO 27001, and HIPAA knowledge-retention audits.

Official Round 2 Submission Certification:
I hereby certify as Team Leader of **TEAM DOMINATOR** that this submission represents a fully functional, original software product developed for Round 2 of the CodersInPune Hackathon. All source code is committed to our public GitHub repository and the working live link operates autonomously without authentication.

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